Synoptic CB: Porewater DOC, TN

June 2022 Samples

2026-06-17

Contents

##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

```
## Setup Information
```

```
###### Run information - PLEASE CHANGE
  Date_Run = "06/15/22"  #Date that instrument was run
  Run_by = "Stephanie J. Wilson"  #Instrument user
  Script_run_by = "Stephanie J. Wilson" #Code user
  run_notes = "Standard Curve assessed manually on the instrument. Some sample IDs are missing from meta
  samples <- c("GCrew", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names

  samples_pattern <- paste(samples, collapse = "|")
    #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
  chks_name = "CKSTD"  #what did you name your check standards?

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
    raw_file_name = c("Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202206_1.txt" , "Raw Data/TOCTN_COMPASS_Sync
```

```r
#file path and name for raw all peaks file
   # raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt"

#file path and name of processed data file
    processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202206.csv"


###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2022.csv"

#qaqc log file path for this year
  #Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAQClog_2022.csv"
```

##Set Up Code

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data Functions

0.2 Import Sample Data

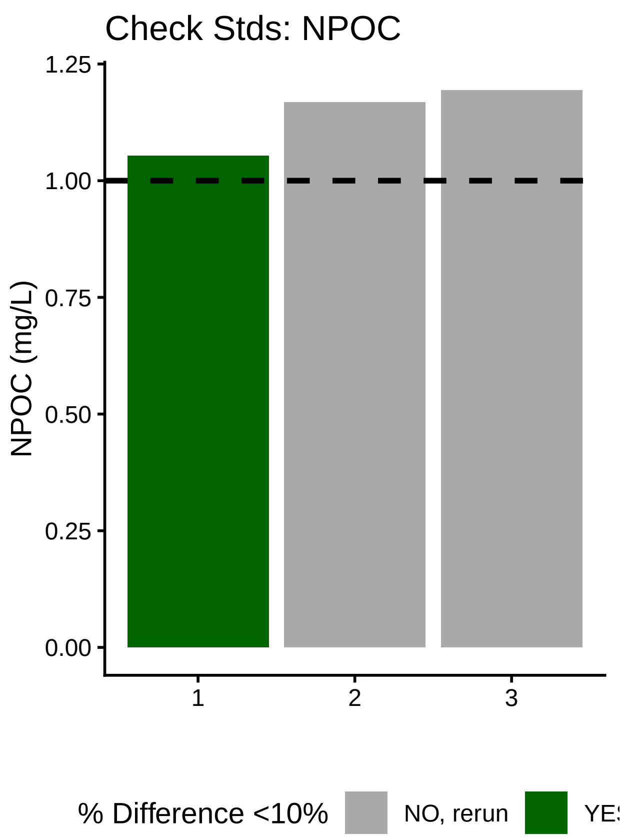
##Fix sample naming

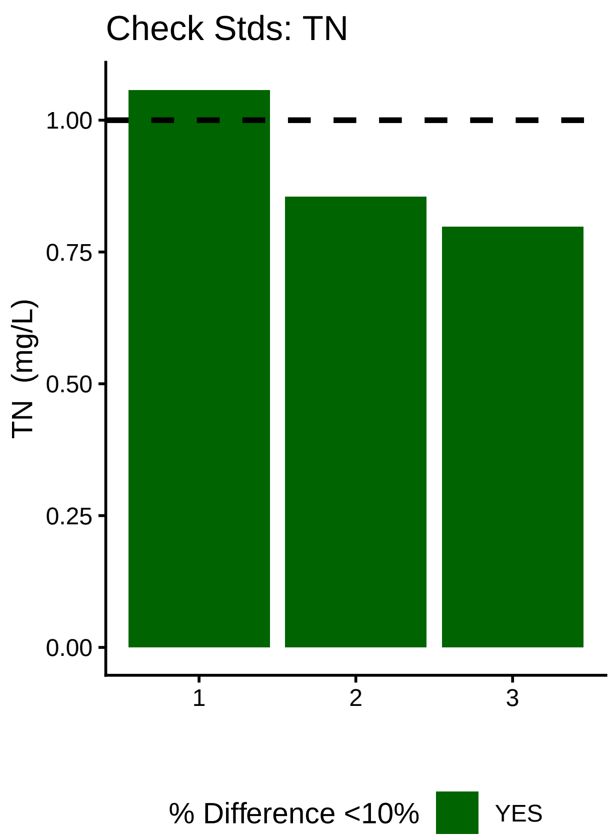
```r
#metadata: 202207_GWI_UP_LysB_20cm
#samples: MSM_WC_PW_MonMon_DOC_20220607_SipA_20cm
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("MSM_UP_PW_MonMon_DOC_20220607", "20220607_MSM_UP", sample_name)) %>%
  mutate(sample_name = gsub("MSM_WC_PW_MonMon_DOC_20220607", "20220607_MSM_WC", sample_name))

##just need the sample year and month
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("20220607", "202206", sample_name))

##fix GCW designation
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("GCrew", "GCW", sample_name))
```

0.3 Assessing Standard Curves - assessed manually on the instrument

0.4 Assess Check Standards

```
## Assess the Check Standards

## New names:
## New names:
## * `` -> `...14`

## [1] "Carbon Check Standard RSD within Range"

## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```

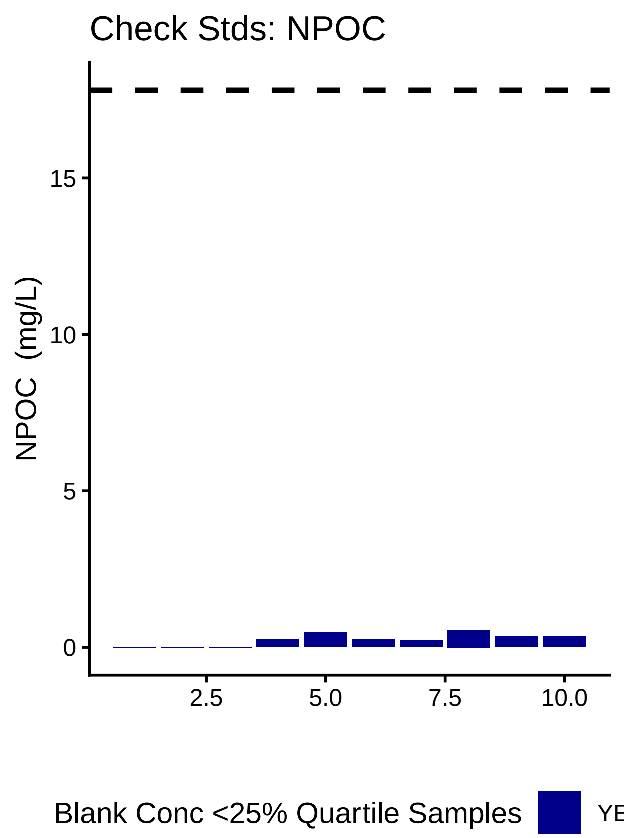


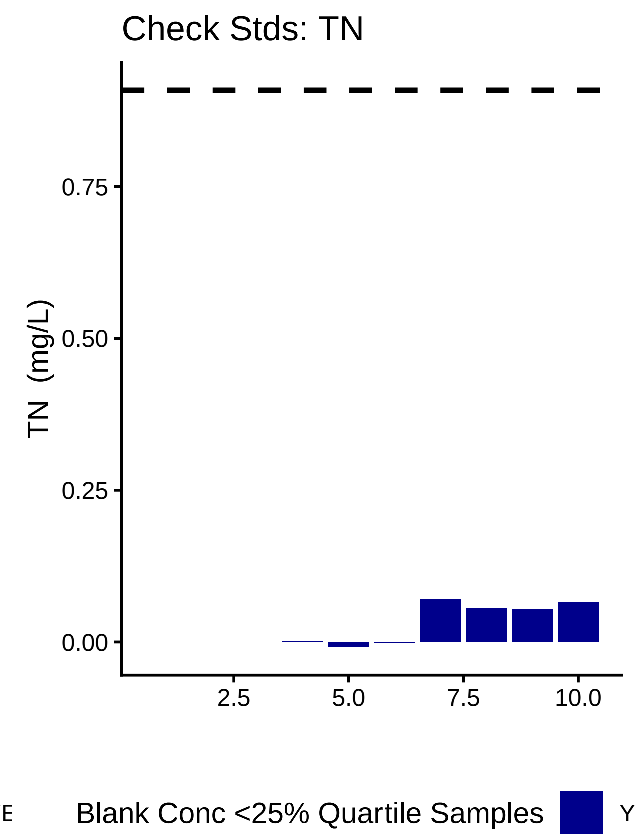



```
## [1] "<50% of Carbon Check Standards are within range of expected concentration - REASSESS"

## [1] ">50% of Nitrogen Check Standards are within range of expected concentration"
```

0.5 Assess Blanks

```
## Assess Blanks

## New names:
## New names:
## * `` -> `...14`

## [1] ">50% of Carbon Blank concentrations are lower 25% quartile of samples"

## [1] ">50% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



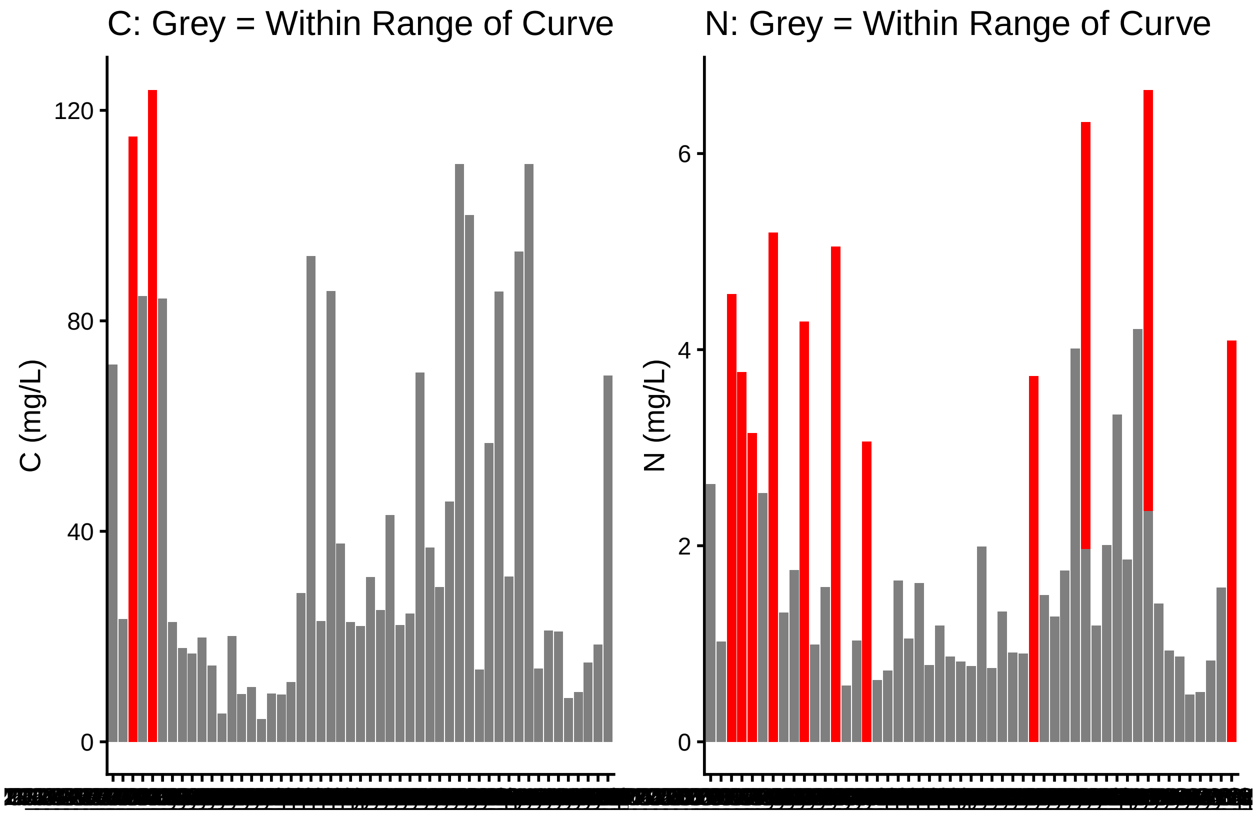



```
## carbon blanks:

## [1] 0.25147

## nitrogen blanks:

## [1] 0.023999
```

0.6 Assess Duplicates - no duplicates ran

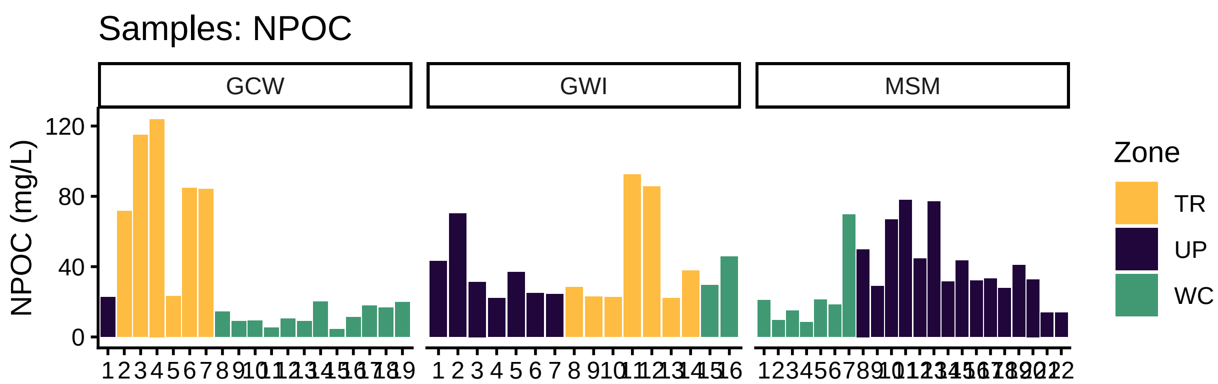
0.7 Sample Flagging - Are samples Within the range of the curve?

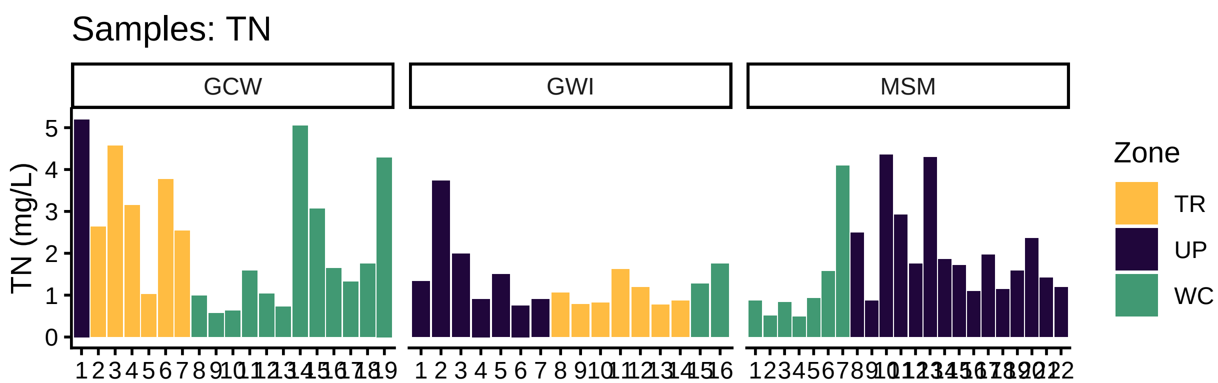
```
## Sample Flagging
```



0.8 Visualize Data by Plot

```
## Visualize Data
```





0.9 Convert data from mg/L to uMoles/L

0.10 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata

## Some sample IDs are missing from metadata.

##  [1] "202206_MSM_WC_SipA_20cm" "202206_MSM_WC_SipB_20cm"
##  [3] "202206_MSM_WC_SipC_10cm" "202206_MSM_WC_SipB_10cm"
##  [5] "202206_MSM_WC_SipA_10cm" "202206_MSM_WC_SipC_20cm"
##  [7] "202206_MSM_WC_SipC_45cm" "202206_MSM_UP_LysB_45cm"
##  [9] "202206_MSM_UP_LysC_45cm" "202206_GWI_TR_LysB_45cm"
## [11] "202206_GWI_WC_SipA_20cm" "202206_GWI_WC_SipC_20cm"
## [13] "202206_GCW_TR_LysB_10cm" "202206_GCW_TR_LysC_10cm"
## [15] "202206_GCW_TR_LysB_20cm" "202206_GCW_TR_LysC_20cm"
## [17] "202206_GCW_WC_SipA_10cm" "202206_GCW_WC_SipB_10cm"
## [19] "202206_GCW_WC_SipC_10cm" "202206_GCW_WC_SipA_20cm"
## [21] "202206_GCW_WC_SipB_20cm" "202206_GCW_WC_SipC_20cm"
## [23] "202206_GCW_WC_SipA_45cm" "202206_GCW_WC_SipB_45cm"
## [25] "202206_GCW_WC_SipC_45cm"
```

0.11 Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 21
##   Project      Region Site  Zone  Replicate Depth_cm Sample_ID  Year Month   Day
##   <chr>        <chr>  <chr> <chr> <chr>     <chr>    <chr>     <int> <int> <int>
## 1 COMPASS: Sy~ CB     MSM   WC    SipA      20       202206_M~    NA    NA    NA
## 2 COMPASS: Sy~ CB     MSM   WC    SipB      20       202206_M~    NA    NA    NA
## 3 COMPASS: Sy~ CB     MSM   WC    SipC      10       202206_M~    NA    NA    NA
## 4 COMPASS: Sy~ CB     MSM   WC    SipB      10       202206_M~    NA    NA    NA
## 5 COMPASS: Sy~ CB     MSM   WC    SipA      10       202206_M~    NA    NA    NA
## 6 COMPASS: Sy~ CB     MSM   WC    SipC      20       202206_M~    NA    NA    NA
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

#end